Pandas

Data Analysis Task

Apply your Pandas knowledge to explore and analyze a real-world dataset.

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Task Overview

You've learned the tools how to use them. In this task, you'll work with the real Titanic passenger dataset the way a data analyst would, figure out what you have, fix what's broken, and dig until you understand what factors determined who survived..

Your Dataset

Dataset URL: Titanic-dataset

Task Sections

Step 1

Load the Data

Your starting point. Get the data into your environment and take your first look at what you're working with.

Step 2

Data Cleaning

Real data is messy. Think carefully about what needs fixing before drawing any conclusions.

Step 3

Exploratory Data Analysis (EDA)

Dig in. What patterns, distributions, or relationships can you find? Let the data surprise you.

Step 4

Analysis & Insights

Go deeper. Ask interesting questions and answer them with Pandas. Visualizations are your friend here.

Step 5

Conclusion

What did you find? Summarize your key takeaways and what they might mean in the real world.

Submission

Your submission must include:

Deliverable	Description
Python Code	A clean, well-commented .py file or Jupyter Notebook (.ipynb) covering all sections.
Written Report	A PDF or Word document discussing your findings, methodology, and conclusions.
Presentation (Optional)	A slide deck summarizing your work is welcome as a bonus deliverable.

A Note on Grading

The quality of your work matters more than the quantity of code. Focus on understanding your data deeply, cleaning it thoughtfully, and communicating your findings clearly.

Good luck! — Marwan Abbas | Vice Head of AI Sub-Team | IEEE SSCS